

Video transcript

Threat and vulnerability assessment

Philosopher Sun Tzu once said, “Know your enemy and know yourself, and you need not fear the outcome of 100 battles.” For effective cybersecurity, you need to understand potential attackers, their motivations, and what tools or methods they might use. When you understand your attackers, you can then assess your defenses clearly and strengthen them in the right ways.

There are many threats to learn about. With the ever-increasing persistence of attackers and the endless amounts of information available online, threat intelligence can be overwhelming. Threat intelligence is defined as information about threats and attackers that can be used to create effective defenses. You can’t assess vulnerability accurately without understanding threats clearly. Threat intelligence usually falls into three categories: strategic intelligence, tactical intelligence, and operational intelligence. Strategic intelligence helps you understand the current and common threat trends and adversarial motives at a high level. Tactical intelligence details how malware works and the specific technical steps others have taken to avoid breaches. Tactical intelligence is useful for cybersecurity professionals who implement defenses. Operational intelligence is even more detailed information that targets very specific threats. It identifies where a threat comes from, what it does, how to prevent it, and how to respond to it if you are affected.

After you gather threat intelligence, what do you do with it? The threat intelligence lifecycle involves five major components: determining intelligence requirements, intelligence collection, intelligence analysis, dissemination of findings, and feedback. These steps ensure that the organization gets needed information and uses it wisely. And, of course, this practice enables consistently effective vulnerability awareness. Keeping this cycle going ensures that threat data is up to date and relevant. Threat intelligence has a direct impact on all aspects of an organization, so the results should be used to guide risk assessments, policies and procedures, network infrastructure, and security controls.

While threat intelligence involves gathering information about potential external threats, a threat assessment is more specific to an organization. It’s a process of understanding the strengths and weaknesses in an organization’s particular systems and network infrastructure. Specifically, vulnerability assessment involves scanning networks, hosts, applications, databases, and wireless assets. A vulnerability assessment defines vulnerabilities, then identifies, classifies, and prioritizes them so that they can be addressed. So, know your enemies and their potential threats. And, know yourself and your vulnerabilities and mitigate those risks. In this way, you need not fear attacks. Using your knowledge of vulnerability assessment, you can help ensure the confidentiality, integrity, and availability of your network.